

Generation Interconnection System Impact Study Report

For Queue Project Q19-1230

Farrell 138 kV

May 25, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1230. The Interconnection Customer proposes a 105.5 MW solar photovoltaic facility interconnecting at the Farrell 138 kV substation in the Virginia area. The request was received on June 18, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 321 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,621,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Old Chapel / Hp Hoo Ct 138kV line
Contingency: Santa Rosa Energy Ctr / Balch 138kV
Loading: 109.7% (Rating: 291 MVA)
DFAX: 6.7%
Pre-project loading: 107.3%
- Facility: Belleair Substation / Wpcp 138kV
Contingency: Santa Rosa Energy Ctr / Balch 138kV
Loading: 103.5% (Rating: 278 MVA)
DFAX: 6.7%
Pre-project loading: 101.0%
- Facility: Santa Rosa Energy Ctr / Monoith Substation 138kV
Contingency: Kearneyville / International Paper Valliant Ok Substation 345kV
Loading: 100.9% (Rating: 252 MVA)
DFAX: 8.1%
Pre-project loading: 97.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Old Chapel / Hp Hoo Ct 138kV line	Reconductor line	\$2,921,000	34% (\$1,006,000)
Belleair Substation / Wpcp 138kV	Reconductor line	\$706,000	34% (\$243,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Santa Rosa Energy Ctr / Monolith 138kV	Reconductor line	\$1,776,000	21% (\$372,000)
	Total allocated to project		\$1,621,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.